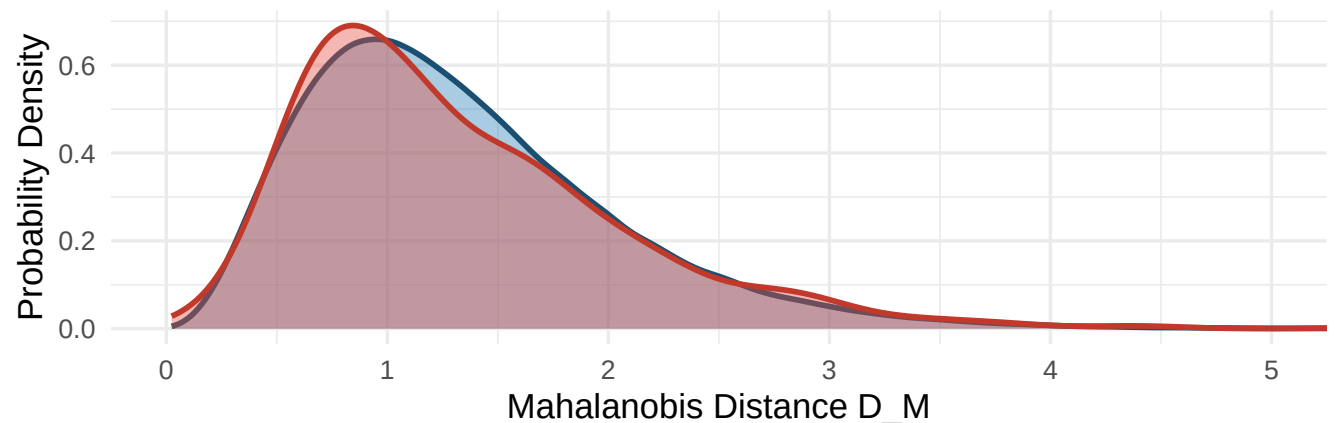


A. Posterior Predictive Check: Empirical vs. Simulated Pause

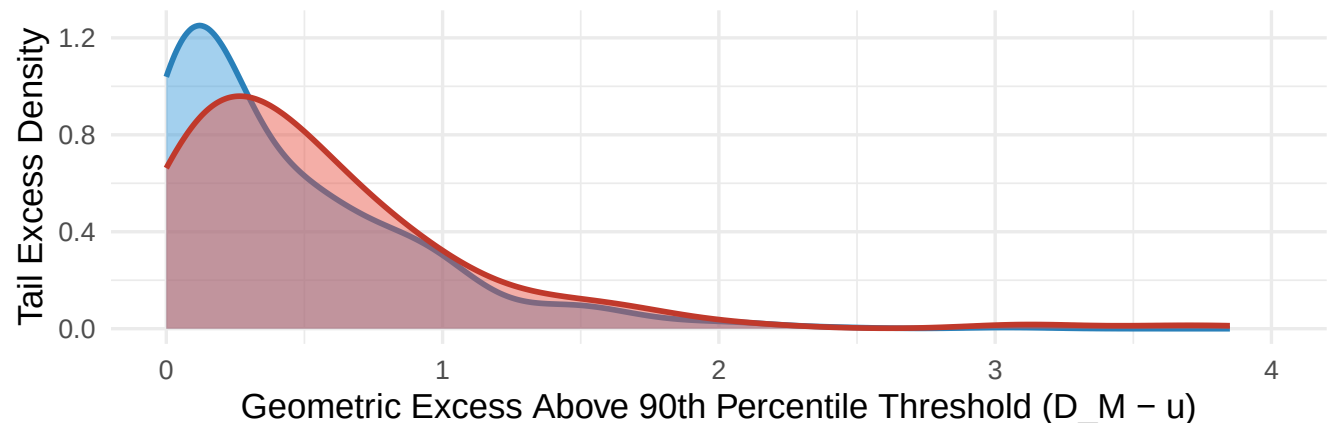
Bayesian Regularized Gamma LMM Accurately Matches the Right-Skewed Head



Type ■ Bayesian HMC Posterior Predictive Simulations ■ Empirical True Pause D_M

B. Extreme Value Theory: Tail Excess Densities (GPD POT)

Pause Events Broaden Tail Dispersion: $\sigma_{\text{pause}} = 0.538$ vs. $\sigma_{\text{null}} = 0$.



Condition ■ Empirical Null Excesses ($u=90\text{th}$) ■ Pause Event Excesses ($u=90\text{th}$)